EXP NO:5 DATA COLLECTION AND INITIAL EXPLORATION

CODE :

```
import pandas as pd
db=pd.read_csv("/content/diabetes.csv")
#Dataset structure and summary statistics
print(db.head())
print(db.describe())
#Visualize the distribution of each feature
import matplotlib.pyplot as plt
import seaborn as sns
db.hist(bins=50,figsize=(20,15))
plt.show()
sns.pairplot(db)
#Pairplot to visualize relationships
plt.show()
```

OUTPUT :

```
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin
BMI  \
0            6      148             72             35        0  33.6
1            1       85             66             29        0  26.6
2            8      183             64              0        0  23.3
```

3 1 89 66 23 94 28.1

4 0 137 40 35 168 43.1

DiabetesPedigreeFunction Age Outcome

0 0.627 50 1

1 0.351 31 0

2 0.672 32 1

3 0.167 21 0

4 2.288 33 1

Pregnancies Glucose BloodPressure SkinThickness Insulin \

count 768.000000 768.000000 768.000000 768.000000 768.000000

mean 3.845052 120.894531 69.105469 20.536458 79.799479

std 3.369578 31.972618 19.355807 15.952218 115.244002

min 0.000000 0.000000 0.000000 0.000000 0.000000

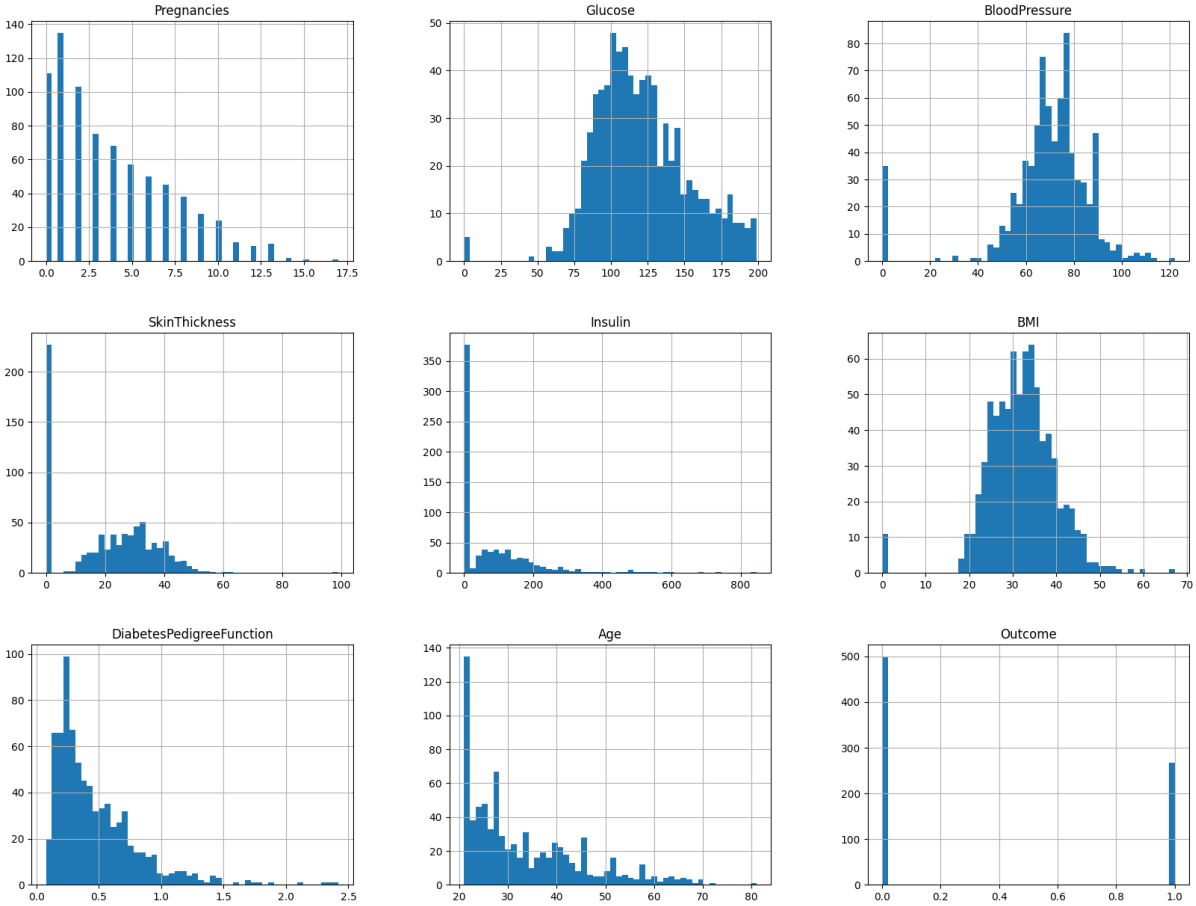
25% 1.000000 99.000000 62.000000 0.000000 0.000000

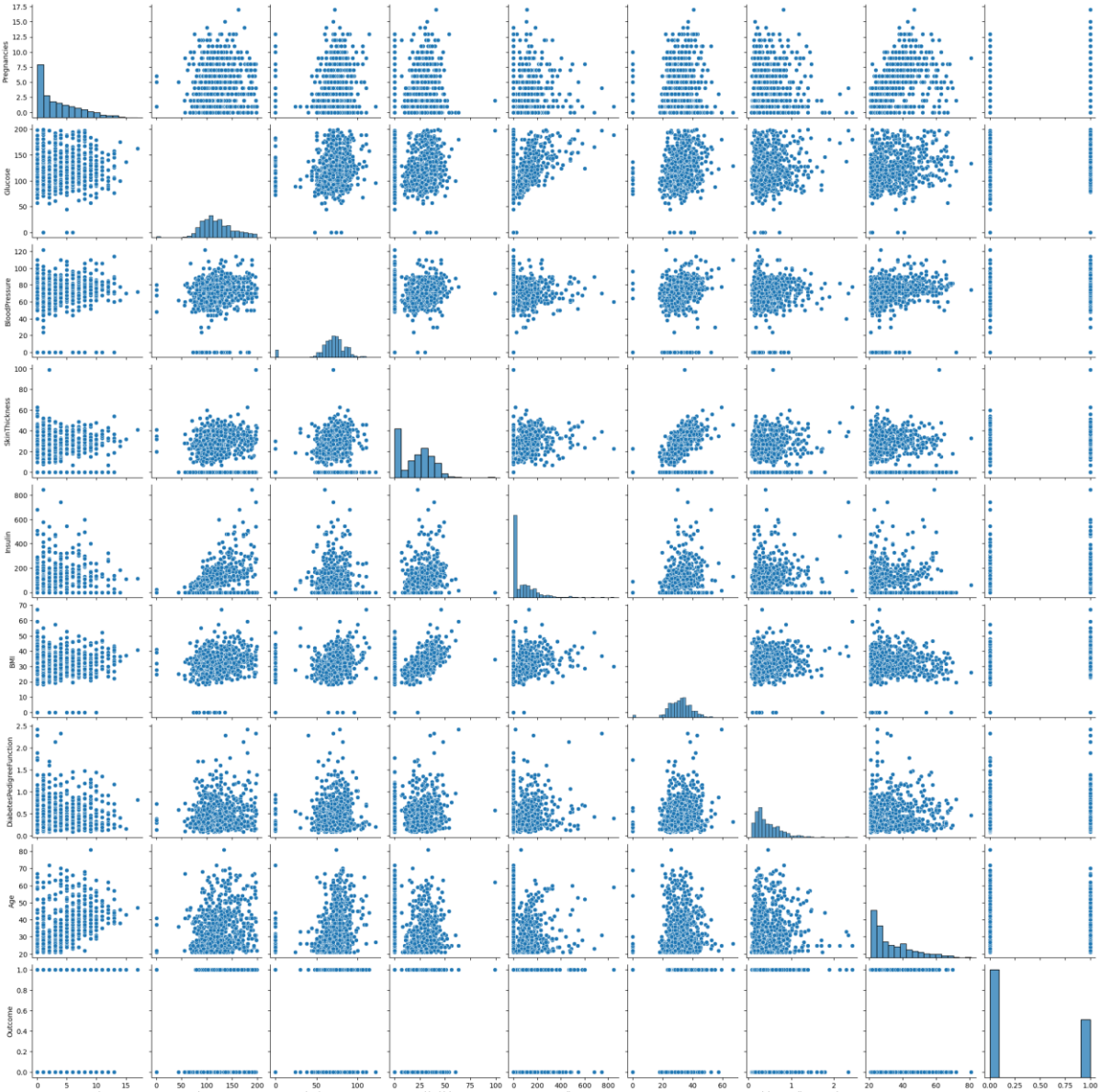
50% 3.000000 117.000000 72.000000 23.000000 30.500000

75% 6.000000 140.250000 80.000000 32.000000 127.250000

max 17.000000 199.000000 122.000000 99.000000 846.000000

	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000
mean	31.992578	0.471876	33.240885	0.348958
std	7.884160	0.331329	11.760232	0.476951
min	0.000000	0.078000	21.000000	0.000000
25%	27.300000	0.243750	24.000000	0.000000
50%	32.000000	0.372500	29.000000	0.000000
75%	36.600000	0.626250	41.000000	1.000000
max	67.100000	2.420000	81.000000	1.000000


Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Outcome


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Glucose
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